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Roll No

AU-702(A) (GS)

B.E. VII Semester Examination, June 2020

Grading System (GS)

Renewable Energy System

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. What is meant by renewable sources? Explain in brief these energy sources with special reference to Indian context.
2. Calculate the zenith and solar altitude angles at 10:00am on 3rd February at a place with latitude of 30°.
3. Define with reference to solar energy :
 - i) Solar radiations
 - ii) Extraterrestrial radiations and
 - iii) Beam radiations
4. Describe working of a Wind Energy Conversion System (WECS) with components.
5. How are geothermal classified? What are the environmental problems associated with geothermal energy?

OR

Draw the P-I and V-I characteristics of fuel cell. Also, derive expression for overall efficiency of a fuel cell.

6. Explain different types of turbines used in small hydro, mini hydro and micro hydro electric plants.
7. Differentiate between VAWT and HAWT.
8. Define the terms:
 - i) Angle of attack
 - ii) Pitch angle
 - iii) Nacelle and
 - iv) Cut-in speed
